

QIAN GORLOS, China

WMO: 509490

Lat: 45.0908N

Lon: 124.8670E

Elev: 446

StdP: 14.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-11.8	-24.9	1.4	-13.4	-21.0	1.8	-7.9	15.5	15.3	13.8	15.4	2.9	220	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	89.3	70.6	86.9	70.7	84.6	69.9	76.6	83.7	75.0	81.8	73.4	80.3	7.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	131.2	81.2	72.8	123.8	79.7	71.1	116.6	78.1	40.6	83.9	39.0	81.7	37.5	80.3	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	16.7	14.5	12.6	DB	-18.8	94.1	7.7	3.1	-24.4	96.3	-28.9	98.1	-33.2	99.8	-38.9	102.0
(5)				WB	-19.2	78.5	7.6	2.2	-24.7	80.1	-29.1	81.3	-33.4	82.6	-38.9	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	4.0	13.4	29.2	47.1	61.0	71.7	75.6	72.5	61.7	45.8	25.4	8.4	
	DBStd	26.41	8.54	10.16	10.14	8.80	7.36	5.90	4.12	4.57	6.55	8.70	11.09	8.95	
	HDD50	5478	1427	1025	648	156	5	0	0	0	3	185	740	1289	
	HDD65	8848	1892	1445	1111	539	169	11	1	5	137	595	1189	1754	
	CDD50	2974	0	0	2	70	347	651	794	698	355	56	1	0	
	CDD65	868	0	0	0	3	46	212	329	238	39	1	0	0	
	CDH74	6850	0	0	0	36	468	1781	2619	1640	295	11	0	0	
	CDH80	1979	0	0	0	4	128	618	785	403	40	1	0	0	
Wind	WSAvg	5.6	4.6	5.5	6.8	7.6	6.9	5.7	5.0	4.6	4.9	5.6	5.8	4.9	
Precipitation	PrecAvg	17.2	0.0	0.1	0.2	0.7	1.3	3.0	5.6	3.6	1.7	0.7	0.2	0.1	
	PrecMax	23.3	0.2	0.4	0.9	5.9	3.3	6.3	9.9	7.9	4.7	2.4	0.6	0.4	
	PrecMin	9.6	0.0	0.0	0.0	0.0	0.2	0.6	2.0	0.9	0.4	0.0	0.0	0.0	
	PrecStd	3.0	0.0	0.1	0.2	1.1	0.9	1.3	1.9	2.0	1.1	0.6	0.2	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	43.8	63.2	78.8	88.1	94.4	92.1	89.1	84.2	75.2	56.9	39.4	
		MCWB	25.5	33.1	46.1	55.0	62.6	68.0	74.9	74.5	67.7	58.0	47.1	34.9	
	2%	DB	24.9	38.6	54.9	72.3	83.6	89.9	89.0	86.5	80.5	68.7	51.4	30.5	
		MCWB	20.8	30.7	40.8	51.6	60.5	67.1	73.7	73.0	65.5	55.4	43.3	26.7	
	5%	DB	20.8	34.1	49.8	67.9	79.3	86.7	86.7	84.3	77.5	63.8	46.3	26.0	
		MCWB	17.3	27.6	38.1	49.7	58.7	67.1	73.3	71.3	63.5	52.3	39.8	22.2	
	10%	DB	17.0	29.5	45.2	63.4	75.0	83.3	84.5	82.1	74.3	59.9	40.6	21.5	
		MCWB	14.0	23.9	34.9	48.0	56.8	66.4	72.5	70.4	61.4	50.1	34.8	18.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	35.0	47.0	58.1	66.8	73.4	79.9	78.0	72.0	61.2	48.9	34.7
			MCDB	30.3	41.8	61.0	72.3	82.0	84.7	87.6	85.9	80.1	68.6	54.9	38.4
		2%	WB	20.9	31.2	42.1	53.9	63.3	71.5	77.4	76.3	68.6	57.2	43.9	27.1
			MCDB	24.7	38.0	53.6	69.4	77.5	83.4	84.5	83.2	76.0	67.2	50.4	30.2
5%		WB	17.4	27.4	38.6	51.1	60.8	70.1	75.9	74.6	66.0	54.5	39.6	22.3	
		MCDB	20.7	34.2	49.0	66.0	74.6	80.6	82.7	80.8	73.9	62.3	45.4	25.9	
10%		WB	14.1	24.0	35.3	48.4	58.6	68.6	74.4	72.8	63.7	51.0	35.1	18.3	
		MCDB	17.0	29.4	44.3	61.5	72.7	78.7	81.3	78.9	71.6	58.8	40.3	21.4	
Mean Daily Temperature Range		5% DB	MDBR	18.0	19.6	18.6	20.4	19.1	16.8	14.4	15.5	19.9	18.8	15.7	16.2
			MCDBR	21.0	22.4	24.9	27.1	25.2	22.1	17.8	18.4	23.6	23.8	20.6	19.3
	MCWBR		17.8	16.7	14.8	13.9	10.4	8.2	7.0	7.7	10.4	13.0	14.2	16.2	
	5% WB	MCDBR	20.9	21.6	23.7	23.7	21.1	17.3	14.2	14.6	18.7	21.1	19.7	19.1	
		MCWBR	17.9	16.4	14.8	13.7	10.2	7.4	7.0	7.7	10.3	13.4	14.6	16.4	
	Clear-Sky Solar Irradiance	taub		0.315	0.335	0.411	0.509	0.582	0.643	0.595	0.525	0.465	0.450	0.368	0.326
taud		2.125	2.109	1.963	1.788	1.687	1.656	1.796	1.949	2.006	1.937	2.067	2.093		
Ebn at Noon		242	262	254	239	226	213	222	233	235	214	216	220		
Edh at Noon		30	37	50	65	75	77	66	55	48	44	32	28		
All-Sky Solar Radiation	RadAvg		674	1020	1378	1645	1826	1796	1689	1590	1411	996	662	548	
	RadStd		64	72	77	121	135	148	156	105	113	62	52	36	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page